geodesic. A straight line in a curved space or curved spacetime. On the Earth's surface the geodesics are the great circles.

gigantic black hole. A black hole that weighs as much as a million Suns, or more. Such holes are thought to inhabit the cores of galaxies and quasars.

global methods. Mathematical techniques, based on a combination of topology and geometry, for analyzing the structure of spacetime.

gravitational cutoff. Oppenheimer's phrase for the formation of a black hole around an imploding star.

gravitational lens. The role of a gravitating body, such as a black hole or a galaxy, to focus light from a distant source by deflecting the light rays; see *light deflection*

gravitational redshift of light. The lengthening of the wavelength of light (the reddening of its color) as it propagates upward through a gravitational field.

gravitational time dilation. The slowing of the flow of time near a gravitating body.

gravitational wave. A ripple of spacetime curvature that travels with the speed of light.

graviton. The particle which, according to wave/particle duality, is associated with gravitational waves.

gyroscope. A rapidly spinning object which holds its spin axis steadily fixed for a very long time.

"hair." Any property that a black hole can radiate away and thus cannot hold on to; for example, a magnetic field or a mountain on its horizon.

hoop conjecture. The conjecture that a black hole forms when and only when a body gets compressed so small that a hoop with the critical circumference can be placed around it and twisted in all directions.

horizon. The surface of a black hole; the point of no return, out of which nothing can emerge. Also called the *absolute horizon* to distinguish it from the *apparent horizon.*

hydrogen bomb. A bomb whose explosive energy comes from the fusion of hydrogen, deuterium, and tritium nuclei to form helium nuclei. See also *superbomb.*

hyperspace. A fictitious flat space in which one imagines pieces of our Universe's curved space as embedded.